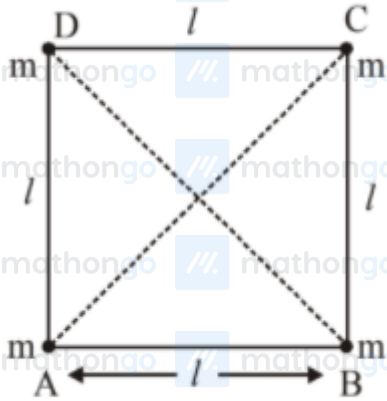


Q1: 16 March (Shift 1) - Single Correct

Four equal masses, m each are placed at the corners of a square of length (l) as shown in the figure. The moment of inertia of the system about an axis passing through A and parallel to DB would be :



- (1) $m/2$
- (2) $2 ml^2$
- (3) $3 ml^2$
- (4) $\sqrt{3} ml^2$

Q2: 16 March (Shift 1) - Numerical

Consider a 20 kg uniform circular disk of radius 0.2 m. It is pin supported at its center and is at rest initially. The disk is acted upon by a constant force $F = 20$ N through a massless string wrapped around its periphery as shown in the figure.



$F = 20\text{N}$

Questions with Answer Keys

MathonGo

Suppose the disk makes n number of revolutions to attain an angular speed of 50rad s^{-1} . The value of n , to the nearest integer, is ____ [Given : In one complete revolution, the disk rotates by 6.28rad]

Q3: 16 March (Shift 2) - Numerical

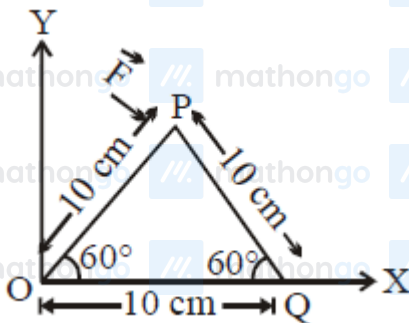
A force $\vec{F} = 4\hat{i} + 3\hat{j} + 4\hat{k}$ is applied on an intersection point of $x = 2$ plane and x -axis. The magnitude of torque of this force about a point $(2,3,4)$ is ____
(Round off to the Nearest Integer)

Q4: 16 March (Shift 2) - Numerical

A solid disc of radius ' a ' and mass ' m ' rolls down without slipping on an inclined plane making an angle θ with the horizontal. The acceleration of the disc will be $\frac{2}{b} g \sin \theta$ where b is _____. (Round off to the Nearest Integer)
(g = acceleration due to gravity)
(θ = angle as shown in figure)

Q5: 17 March (Shift 1) - Single Correct

A triangular plate is shown. A force $\vec{F} = 4\hat{i} - 3\hat{j}$ is applied at point P. The torque at point P with respect to point ' O ' and ' Q ' are :



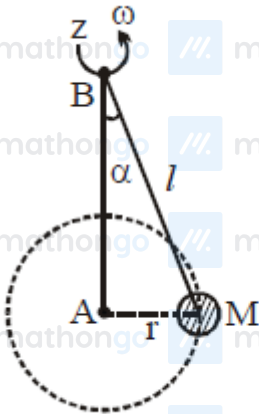
- (1) $-15 - 20\sqrt{3}, 15 - 20\sqrt{3}$
- (2) $15 + 20\sqrt{3}, 15 - 20\sqrt{3}$
- (3) $15 - 20\sqrt{3}, 15 + 20\sqrt{3}$
- (4) $-15 + 20\sqrt{3}, 15 + 20\sqrt{3}$

Questions with Answer Keys

MathonGo

Q6: 17 March (Shift 1) - Single Correct

A mass M hangs on a massless rod of length l which rotates at a constant angular frequency. The mass M moves with steady speed in a circular path of constant radius. Assume that the system is in steady circular motion with constant angular velocity ω . The angular momentum of M about point A is L_A which lies in the positive z direction and the angular momentum of M about B is L_B . The correct statement for this system is :



- (1) L_A and L_B are both constant in magnitude and direction
- (2) L_B is constant in direction with varying magnitude
- (3) L_B is constant, both in magnitude and direction
- (4) L_A is constant, both in magnitude and direction

Q7: 17 March (Shift 1) - Numerical

The angular speed of truck wheel is increased from 900rpm to 2460rpm in 26 seconds. The number of revolutions by the truck engine during this time is _____
(Assuming the acceleration to be uniform).

Q8: 17 March (Shift 1) - Numerical

The following bodies,

- (1) a ring
- (2) a disc
- (3) a solid cylinder

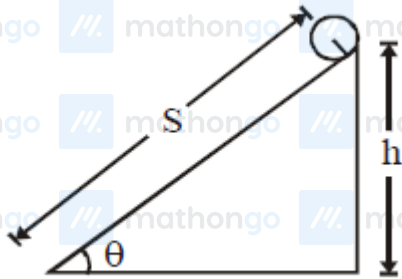
Questions with Answer Keys

MathonGo

(4) a solid sphere,

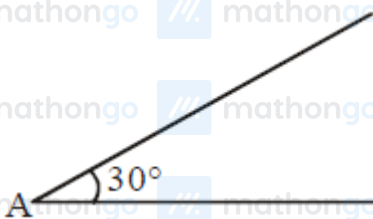
of same mass 'm' and radius 'R' are allowed to roll down without slipping simultaneously from the top of the inclined plane. The body which will reach first at the bottom of the inclined plane is

[Mark the body as per their respective numbering given in the question]



Q9: 17 March (Shift 2) - Single Correct

A sphere of mass 2 kg and radius 0.5 m is rolling with an initial speed of 1 ms^{-1} goes up an inclined plane which makes an angle of 30° with the horizontal plane, without slipping. How low will the sphere take to return to the starting point A ?



(1) 0.60 s

(2) 0.52 s

(3) 0.57 s

(4) 0.80 s

Q10: 18 March (Shift 1) - Single Correct

A thin circular ring of mass M and radius r is rotating about its axis with an angular speed ω . Two particles having mass m each are now attached at diametrically opposite points. The angular speed of the ring will become:

Questions with Answer Keys

MathonGo

(1) $\omega \frac{M}{M+m}$

(2) $\omega \frac{M+2m}{M}$

(3) $\omega \frac{M}{M+2m}$

(4) $\omega \frac{M-2m}{M+2m}$

Q11: 18 March (Shift 2) - Single Correct

Consider a uniform wire of mass M and length L . It is bent into a semicircle. Its moment of inertia about a line perpendicular to the plane of the wire passing through the centre is :

(1) $\frac{1}{4} \frac{ML^2}{\pi^2}$

(2) $\frac{2}{5} \frac{ML^2}{\pi^2}$

(3) $\frac{ML^2}{\pi^2}$

(4) $\frac{1}{2} \frac{ML^2}{\pi^2}$

Q12: 18 March (Shift 2) - Single Correct

A solid cylinder of mass m is wrapped with an inextensible light string and, is placed on a rough inclined plane as shown in the figure. The frictional force acting between the cylinder and the inclined plane is :



[The coefficient of static friction, μ_s , is 0.4]

Rotational Motion

JEE Main 2021 (March)

Questions with Answer Keys

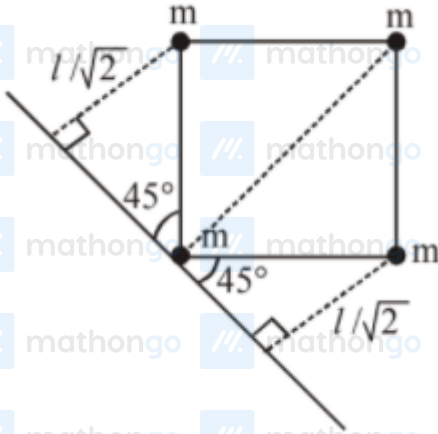
MathonGo

Answer Key**Q1 (3)****Q2 (20)****Q3 (20)****Q4 (3)****Q5 (1)****Q6 (4)****Q7 (728)****Q8 (4)****Q9 (3)****Q10 (3)****Q11 (3)****Q12 (3)**

Q1 (3)

Moment of inertia of point mass

$$= \text{mass} \times (\text{Perpendicular distance from axis})^2$$



Moment of Inertia

$$= m(0)^2 + m(l\sqrt{2})^2 + m\left(\frac{l}{\sqrt{2}}\right)^2 + m\left(\frac{l}{\sqrt{2}}\right)^2$$

$$= 3m/l^2$$

Q2 (20)

$$\alpha = \frac{\tau}{I} = \frac{F.R}{mR^2/2} = \frac{2F}{mR}$$

$$\alpha = \frac{2 \times 200}{20 \times (0.2)} = 10 \text{ rad/s}^2$$

$$\omega^2 = \omega_0^2 + 2\alpha\Delta\theta$$

$$(50)^2 = 0^2 + 2(10)\Delta\theta \Rightarrow \Delta\theta = \frac{2500}{20}$$

$$\Delta\theta = 125 \text{ rad}$$

$$\text{No. of} = \frac{125}{2\pi} \approx 20 \text{ revolution}$$

Q3 (20)

Hints and Solutions

MathonGo

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\vec{r} = (2\hat{i}) - (2\hat{i} + 3\hat{j} + 4\hat{k}) = -3\hat{j} - 4\hat{k}$$

$$\vec{F} = 4\hat{i} + 3\hat{j} + 4\hat{k}$$

$$\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -3 & -4 \\ 4 & 3 & 4 \end{vmatrix}$$

$$= \hat{i}(-12 + 12) - \hat{j}(0 + 16) + \hat{k}(0 + 12)$$

$$= -16\hat{j} + 12\hat{k}$$

$$\therefore |\vec{\tau}| = \sqrt{16^2 + 12^2} = 20$$

Q4 (3)

$$a = \frac{g \sin \theta}{1 + \frac{1}{mR^2}} = \frac{g \sin \theta}{1 + \frac{1}{2}} = \frac{2}{3} g \sin \theta$$

$$b = 3$$

Q5 (1)

$$\vec{F} = 4\hat{i} - 3\hat{j}$$

$$\vec{r}_1 = 5\hat{i} + 5\sqrt{3}\hat{j} \quad \& \quad \vec{r}_2 = -5\hat{i} + 5\sqrt{3}\hat{j}$$

Torque about 'O'

$$\vec{\tau}_0 = \vec{r}_1 \times \vec{F} = (-15 - 20\sqrt{3})\hat{k} = (15 + 20\sqrt{3})(-\hat{k})$$

Torque about 'Q'

$$\vec{\tau}_Q = \vec{r}_2 \times \vec{F} = (-15 + 20\sqrt{3})\hat{k} = (15 - 20\sqrt{3})(-\hat{k})$$

Q6 (4)

$$\text{We know, } \vec{L} = m(\vec{r} \times \vec{v})$$

Now with respect to A, we always get direction

of $\vec{L}$ along the z-axis and also constant magnitude as mvr . But with respect to B, we get constant magnitude but continuously changing direction.

Q7 (728)

$$\text{We know, } \theta = \left(\frac{\omega_1 + \omega_2}{2} \right) t$$

Let number of revolutions be N

$$\therefore 2\pi N = 2\pi \left(\frac{900 + 2460}{60 \times 2} \right) \times 26$$

$$N = 728$$

Q8 (4)

$$Mg \sin \theta R = (mk^2 + mR^2) \alpha$$

$$\alpha = \frac{Rg \sin \theta}{k^2 + R^2} \Rightarrow a = \frac{g \sin \theta}{1 + \frac{k^2}{R^2}}$$

$$t = \sqrt{\frac{2s}{a}} = \sqrt{\frac{2s}{g \sin \theta} \left(1 + \frac{k^2}{R^2} \right)}$$

for least time, k should be least & we know k

is least for solid sphere.

Q9 (3)

$$a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}} = \frac{5}{7} \times \frac{10}{2} = \frac{25}{7}$$

$$t = \frac{2v_0}{a} = \frac{2 \times 1 \times 7}{25} = 0.56$$

Q10 (3)

Using conservation of angular momentum

$$(Mr^2) \omega = (Mr^2 + 2mr^2) \omega'$$

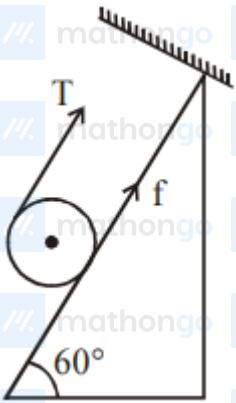
$$\omega' = \frac{M\omega}{M+2m}$$

Q11 (3)

$$\pi r = L \Rightarrow r = \frac{L}{\pi}$$

$$I = Mr^2 = \frac{ML^2}{\pi^2}$$

Q12 (3)



Let's take solid cylinder is in equilibrium

$$T + f = mg \sin 60^\circ \dots (i)$$

$$TR - fR = 0 \dots (ii)$$

Solving we get

$$T = f_{\text{req}} = \frac{mg \sin \theta}{2}$$

But limiting friction < required friction

$$\mu mg \cos 60^\circ < \frac{mg \sin 60^\circ}{2}$$

$\therefore$ Hence cylinder will not remain in equilibrium Hence $f = \text{kinetic}$

$$= \mu_k N$$

$$= \mu_k mg \cos 60^\circ$$

$$= \frac{mg}{5}$$